Digital Images and Image Editing

LIU08106 · BLISS III | RIU08106 · BRIM III

Good images communicate clearly without concealing source information.

Learning outcomes

- Compare scanning and photography as image-capture methods.
- Explain the roles of TIFF, PNG, JPEG and other image formats.
- Prepare and document controlled edits and access derivatives.
- Identify OCR errors and evaluate image quality, rights and accessibility.

Connect technical choices to responsible use in information services.

Where do digital images originate?

- Camera or phone: lens and sensor sample a photographed scene.
- Flatbed scanner: captures a suitable flat original on glass.
- Overhead camera: can reduce pressure on suitable fragile or bound items.
- A screenshot records a display view, not necessarily the original source data.

Source and handling conditions determine the capture method.

Start with audience and material

- Identify whether the item is a photograph, map, page, drawing or fragile object.
- Identify the smallest important detail: handwriting, seal, signature or fine printed characters.
- Define the intended preservation and access outputs.
- Check rights, handling restrictions, identifiers and required documentation.

Work backwards from the material and intended use.

Capture quality: more than resolution

- Adequate pixels cannot correct a badly focused image.
- Check lighting, glare, shadow, camera movement and page curvature.
- Include complete source edges when preparing a managed master.
- Compare the capture against the source for legibility and fidelity.

Technical image quality is multidimensional.

Resolution example

- An item measures 5 inches across and is captured at 400 pixels per inch.
- Estimated pixel width = 5 × 400 = 2,000 pixels.
- The suitable sampling density depends on content and the chosen quality target.
- Do not treat one PPI number as mandatory for every document.

Use source detail and published guidance to set capture requirements.

Handling fragile collections

- Never force a delicate bound item onto a scanner glass.
- Assess bindings, attachments, pages and signs of damage before capture.
- Use suitable supports and work under an approved handling procedure.
- Stop if an action risks damaging an original.

Preservation includes protecting the physical object during digitisation.

What does an image file format do?

- It specifies how image data and related information are organised for storage and reading.
- Formats differ in supported pixel data, metadata, compression and software compatibility.
- The extension .tif or .jpg is not proof of authenticity or quality.
- Choose both capture and delivery formats for a documented reason.

Separate the format label from the actual file properties.

JPEG in practice

- Ordinary JPEG uses lossy compression and often produces small photographs.
- Suitable for many access images, catalogue thumbnails and online galleries.
- Strong compression can add block-like artefacts or halos around letters.
- Repeated saving at lossy settings can further degrade source pixels.

Useful for access, but protect fine-text masters separately.

PNG in practice

- PNG uses lossless image compression and may support transparency.
- Often suitable for diagrams, screenshots, typography and some access derivatives.
- Photographic PNG files may be larger than comparable JPEG access files.
- Lossless compression protects the encoded pixels, not necessarily their provenance.

Match the image type and the delivery requirement.

TIFF in heritage workflows

- TIFF is a flexible container often used for managed still-image capture.
- It can support high-bit-depth pixel data and various compression methods.
- A TIFF may be uncompressed, losslessly compressed or encoded differently.
- Document pixel dimensions, bit depth, colour profile and compression.

“TIFF” alone does not describe every relevant property.

Other image representations

- SVG suits scalable icons, line diagrams and graphical interfaces.
- RAW may retain camera-origin sensor information in some workflows.
- WebP and AVIF can provide efficient online delivery where supported.
- Use compatibility, openness, maintenance and local policy to guide selection.

No single format is ideal for every image and purpose.

Three important file roles

- Master: controlled, high-quality source representation protected from routine edits.
- Working copy: editable copy for experimentation and corrections.
- Access derivative: smaller or adapted file made for a defined user/channel.
- Maintain clear identifiers and links among all versions.

Preserve the source while designing usable access.

Proposed file naming scheme

- DEMO-003_master.tif identifies the controlled original capture.
- DEMO-003_access_1600.jpg identifies a resized web version.
- DEMO-003_thumb.jpg identifies a small navigation image.
- DEMO-003_editlog.txt records the reason for transformations.

Names aid organisation; preservation also requires documentation and controls.

Cropping: reveal and conceal

- Cropping removes visible edges or regions from the delivered frame.
- A crop can improve a focused thumbnail, but may hide notes, borders and dates.
- Retain the full capture in the source/master file.
- Label a cropped derivative accurately and document omitted context.

A crop is an interpretation, not a replacement for the complete original.

Rotate, straighten and perspective

- Rotation helps correct a sideways page; straightening fixes small skew.
- Perspective adjustment may help read a camera-captured page.
- Transformations interpolate pixels and can change detail.
- Compare with source and record significant geometry changes.

Correct presentation without implying the geometry was original.

Resize without distortion

- Downsampling reduces pixel dimensions for a smaller output.
- Upsampling estimates new pixel values and cannot recreate lost capture detail.
- Preserve the original aspect ratio for faithful proportions.
- Example: 2400 × 1600 → 1200 × 800, not 1200 × 1200.

A clear access copy is better than an oversized distorted image.

Brightness and contrast

- Moderate adjustments may aid legibility of faint handwriting.
- Over-brightening can clip highlights; heavy contrast can erase subtle tones.
- Compare the result with the unmodified capture.
- Record the adjustments in an edit log or processing manifest.

Improved appearance does not necessarily mean improved evidential fidelity.

Sharpening and noise reduction

- Sharpening can increase edge contrast, but overuse creates artificial halos.
- Noise reduction can smooth distracting speckles but remove faint strokes.
- Neither algorithm can prove what the source originally said.
- Inspect critical details at 100% and ask whether meaning was altered.

Every enhancement creates a trade-off.

Original, derivative and edit log

- Record input filename, output filename, date, operator and purpose.
- Document cropping area, dimensions, format and notable filter parameters.
- Protect the source image from accidental overwrite.
- Keep enough context to reproduce or explain the displayed access copy.

Transparent processing supports trust and future reuse.

Lossless versus lossy compression

- Lossless decoding restores exactly the data encoded before compression.
- Lossy methods discard some information for smaller sizes.
- JPEG is normally lossy; PNG is normally lossless; TIFF has multiple configurations.
- Report real dimensions, compression settings and actual file sizes.

A small file is not automatically a better file.

Worked uncompressed-size estimate

- Image: 1200 × 800 pixels, 24 bits per pixel (3 bytes/pixel).
- Pixel payload = 1200 × 800 × 3 = 2,880,000 bytes.
- Approximate payload = 2,880,000 ÷ 1,048,576 ≈ 2.75 MiB.
- Container metadata and compression change the actual saved file size.

Do not confuse raw pixel payload with the disk size of a JPEG or PNG.

OCR: pixels become searchable text

- OCR software analyses character shapes in a scanned image.
- Recognised text can support searching, indexing and accessible reading.
- Recognition accuracy changes with source condition, language and layout.
- Image-only scanning does not itself make a page text-searchable.

Treat OCR as a proposed transcription that needs verification.

OCR correction demonstration

- Actual source wording: “LIBRARY RECORDS 1984”.
- Illustrative OCR output: “L1BRARY REC0RDS 1984”.
- The software confused the letter I with numeral 1 and O with numeral 0.
- Compare against the page and record uncertainty rather than inventing corrections.

Seemingly minor OCR errors can cause failed searches or false facts.

OCR for Kiswahili and mixed content

- Choose or test supported recognition language models.
- Inspect local names, punctuation, numbers and bilingual headings manually.
- Handwriting, uneven pages and historical typefaces may reduce accuracy.
- Keep the page image and distinguish verified text from uncertain readings.

Language-sensitive validation makes a collection more discoverable.

Metadata and quality control

- Identify the item, capture date, method, responsible person and technical settings.
- Check completeness, sequence, legibility, colour, orientation and file integrity.
- Record meaningful processing steps, especially for access derivatives.
- Link description, image, OCR text and rights information.

Successful digitisation is a managed workflow, not a button press.

Accessible images and captions

- Use meaningful alternative text for informative webpage images.
- Provide longer descriptions or transcriptions for complex documents.
- Identify the original item and clearly label reconstructed or edited displays.
- Test readability on mobile screens and slow connections.

Users need information, not simply attractive pixels.

BLISS example: library orientation

- Capture or select an authorised photograph of an information desk.
- Make a web-sized access copy and legible labelled thumbnail.
- Link the image to practical instructions and catalogue information.
- Supply a useful caption, alternative text and source attribution.

Images should help users complete library tasks.

BRIM example: archival register

- Assess physical handling and retain a complete capture of each page.
- Record provenance, page order and appropriate identifiers.
- Make a separate reading copy, tracking contrast or crop changes.
- Check OCR results and do not conceal marginal notes or doubtful readings.

Access must not compromise provenance and authenticity.

Practical workflow: edit and document

- Download the fictional sample master from the course site.
- Duplicate it and perform a justified crop or resize in an editor.
- Export an access derivative; preserve the sample source unchanged.
- Check visual detail, write an edit log and compare file dimensions and size.

Your report must explain what changed and why.

Unit recap and self-test

- Explain why capture quality is not measured by megapixels alone.
- Compare TIFF, PNG and JPEG without calling any one universally “best”.
- Explain the relationship among master, working copy and access derivative.
- Identify the limits of cropping, brightness enhancement and OCR.

The same principles of faithful capture, access and documentation apply to other media.